

# LAB REPORT

11th March 2025



## Disk Image Analysis Practical

Name : Sarthak Das

Session : 11th February – 22nd April 2025

For the class of Computer Forensic, taught by

**Drabanti Boral**

# Contents

|          |                                           |          |
|----------|-------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                       | <b>2</b> |
| 1.1      | Objectives . . . . .                      | 2        |
| 1.2      | System Information . . . . .              | 2        |
| 1.3      | Software Used . . . . .                   | 2        |
| <b>2</b> | <b>Objective 1: Mounting Disk Image</b>   | <b>3</b> |
| 2.1      | Methodology: Exterro FTK Imager . . . . . | 3        |
| <b>3</b> | <b>Objective 2: Analyzing Disk Image</b>  | <b>5</b> |
| 3.1      | Methodology: Exterro FTK Imager . . . . . | 5        |
| 3.2      | Methodology: Autopsy . . . . .            | 7        |
| 3.3      | Analysis: Autopsy . . . . .               | 10       |

# 1 Introduction

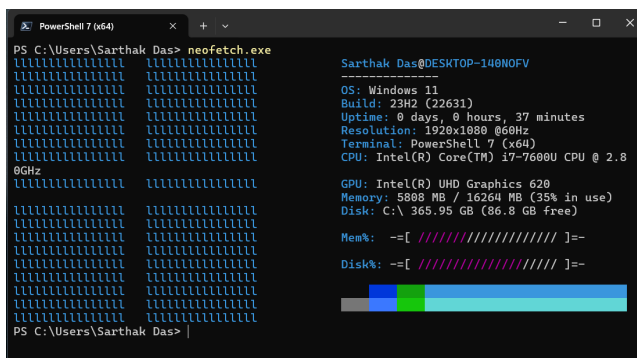
## 1.1 Objectives

This report is a documentation of my lab work as assigned on 6th March, 2025. The primary objectives of this lab are two-fold:

- Mounting a disk image file.
- Analyzing the disk image file.

## 1.2 System Information

For the purposes of this lab, a Lenovo ThinkPad X270 was used, having the following system specifications: Intel Core i7-7600U CPU @ 2.80GHz with Intel UHD Graphics 620, 16GB RAM, 1TB SSD running operating system Windows 11 23H2 (Build 22631).



```
PS C:\Users\Sarthak Das> neofetch.exe

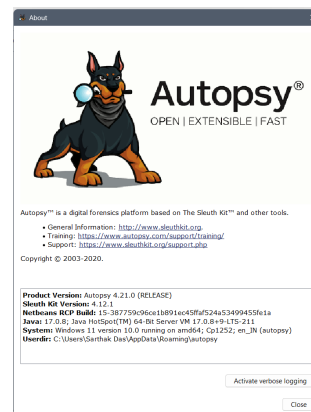
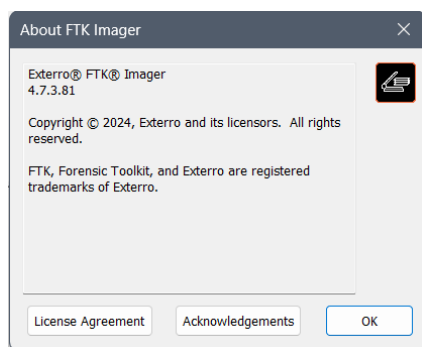
          Sarthak Das@DESKTOP-148NOFV
          OS: Windows 11
          Build: 23H2 (22631)
          Uptime: 0 days, 0 hours, 37 minutes
          Resolution: 1920x1080 @60Hz
          Terminal: PowerShell 7 (x64)
          CPU: Intel(R) Core(TM) i7-7600U CPU @ 2.8
          GHz
          GPU: Intel(R) UHD Graphics 620
          Memory: 5888 MB / 16264 MB (35% in use)
          Disk: C:\ 365.95 GB (86.8 GB free)

          Mem%:  ~=[  //////////////// ]=~
          Disk%: ~=[  //////////////// ]=~

          PS C:\Users\Sarthak Das> |
```

## 1.3 Software Used

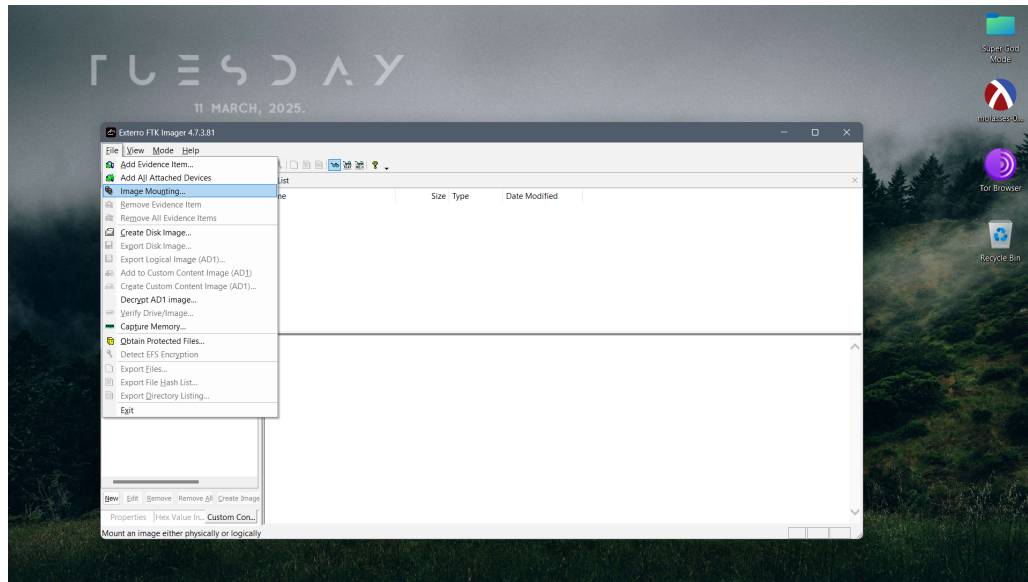
The software used for this disk image analysis practical are Exterro FTK Imager version 4.7.3.81 and Autopsy 4.2.0 (RELEASE).



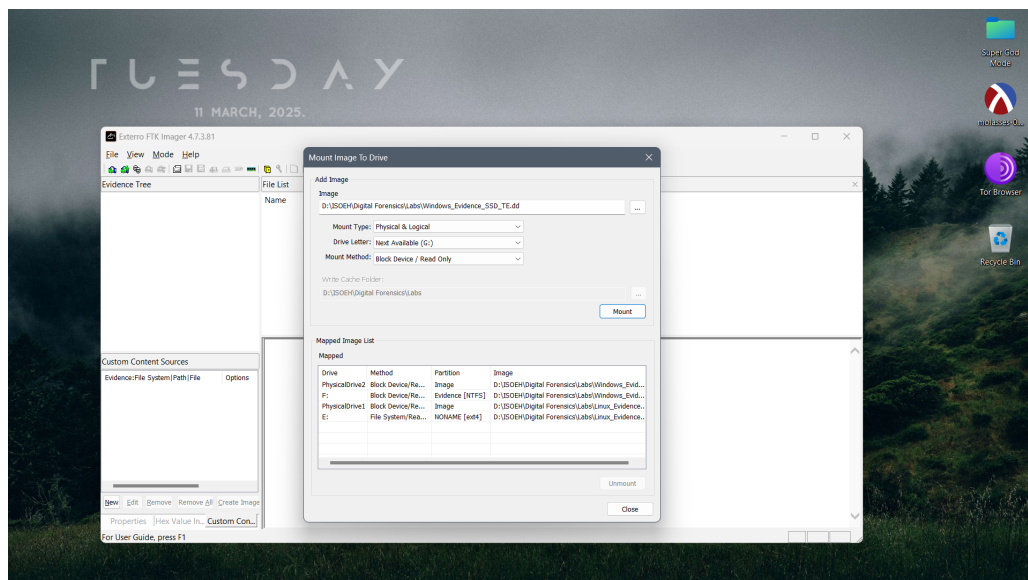
## 2 Objective 1: Mounting Disk Image

### 2.1 Methodology: Exterro FTK Imager

We begin by opening Exterro FTK Imager, and selecting **File** > *Image Mounting*.

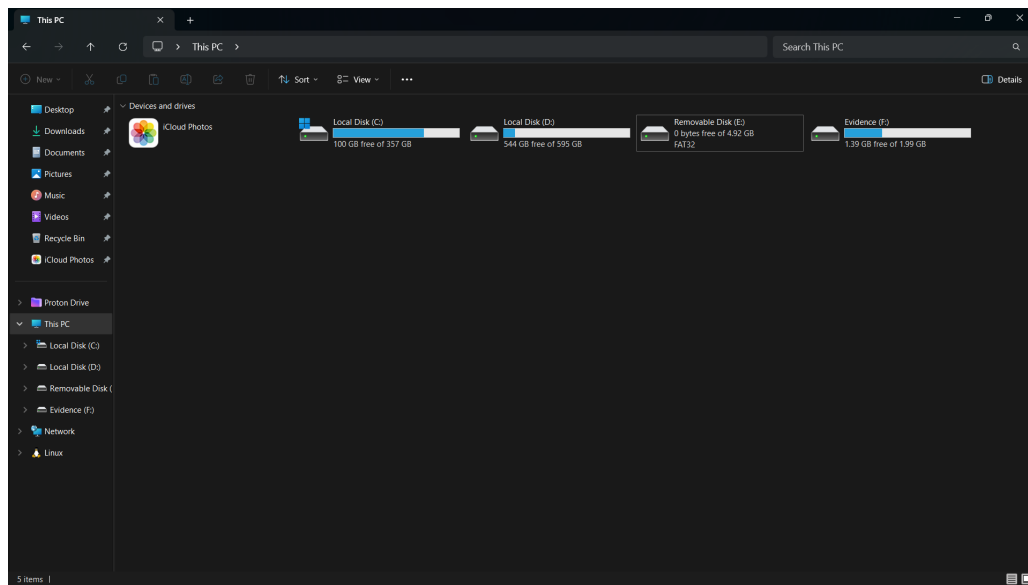


Next, we select the disk image file we wish to mount and click **Mount**. For the purposes of this lab, we were provided with two evidence disk image drives – one Linux (4.92GB ext4) and one Windows (1.99GB NTFS).





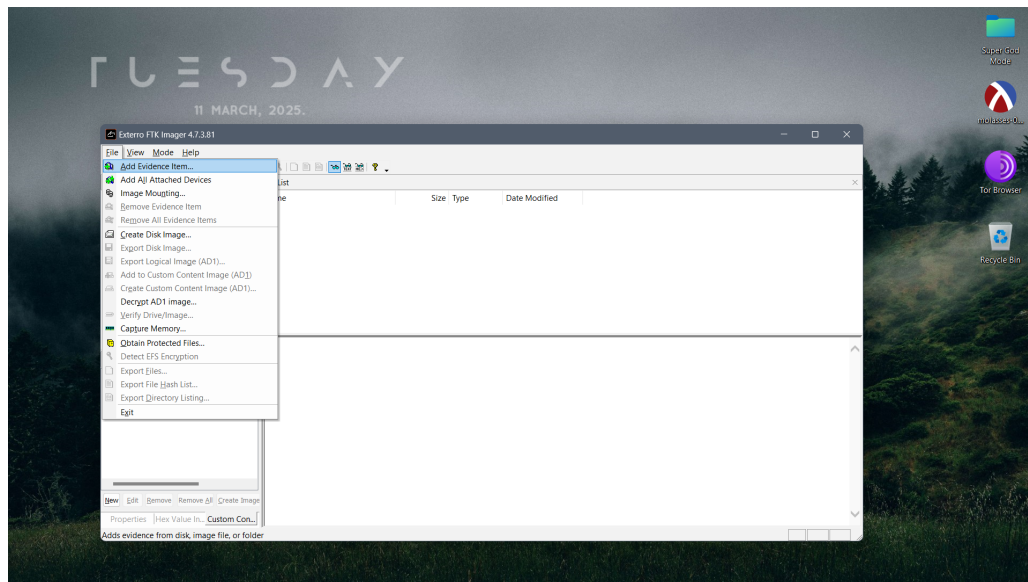
Upon mounting, we can access the disk images using **File Explorer**. Note that Windows does not natively support the ext4 file format, hence the Linux image will be interpreted as a FAT32 volume, hence proper analysis cannot be done without the use of specialized software.



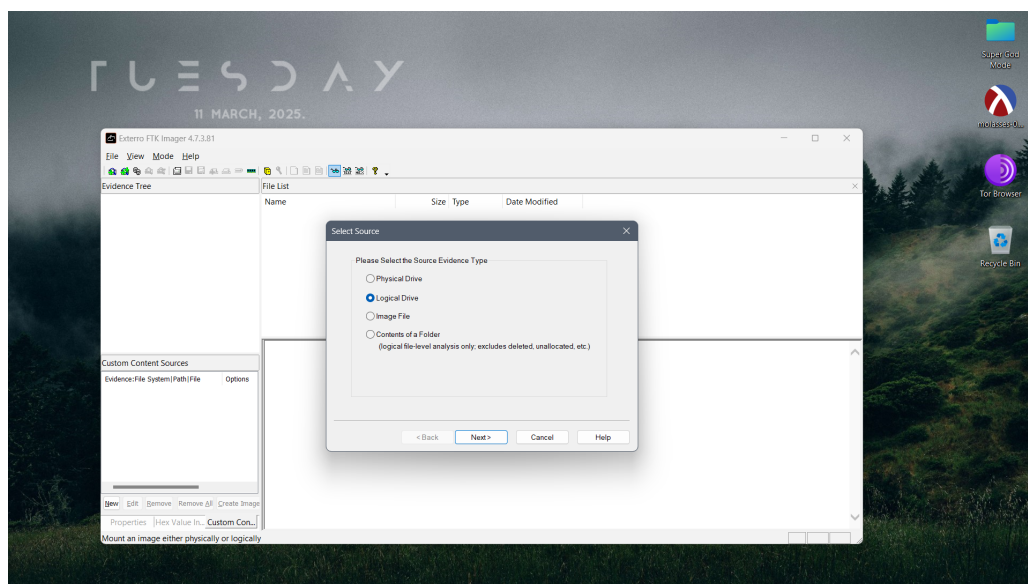
## 3 Objective 2: Analyzing Disk Image

### 3.1 Methodology: Exterro FTK Imager

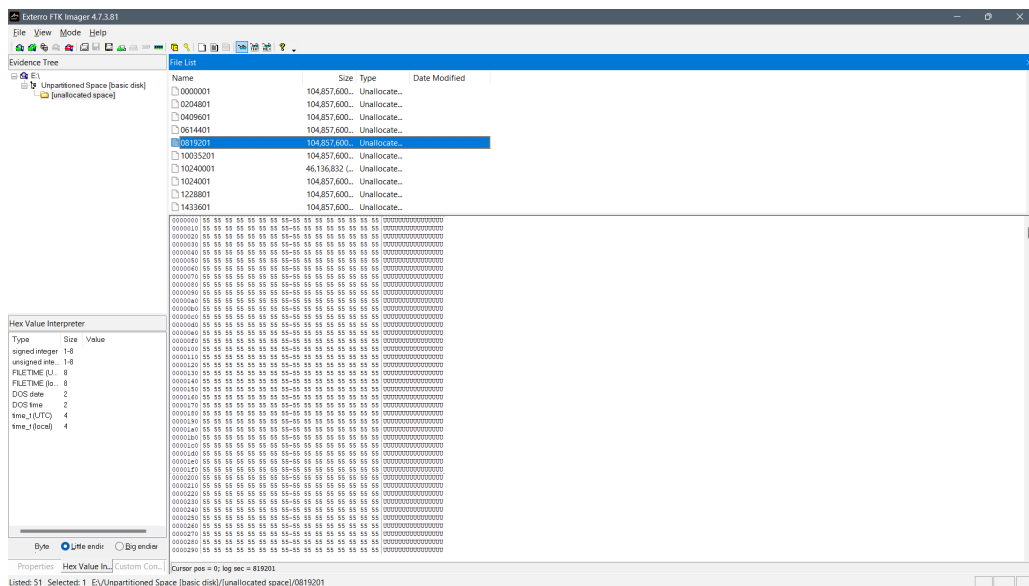
We begin by opening Exterro FTK Imager, and selecting **File > Add Evidence Item**.



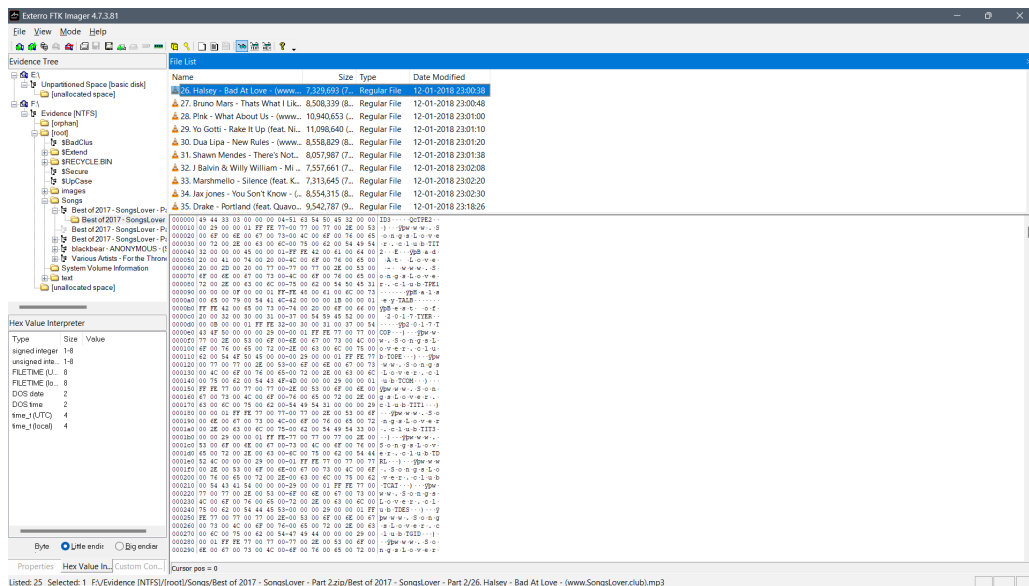
In our case, we select the “Logical Drive” option and choose the disk image (Linux) we have already mounted into the file system.



We can now browse the **evidence tree**, and analyze the properties, hex values and hash lists of the files.

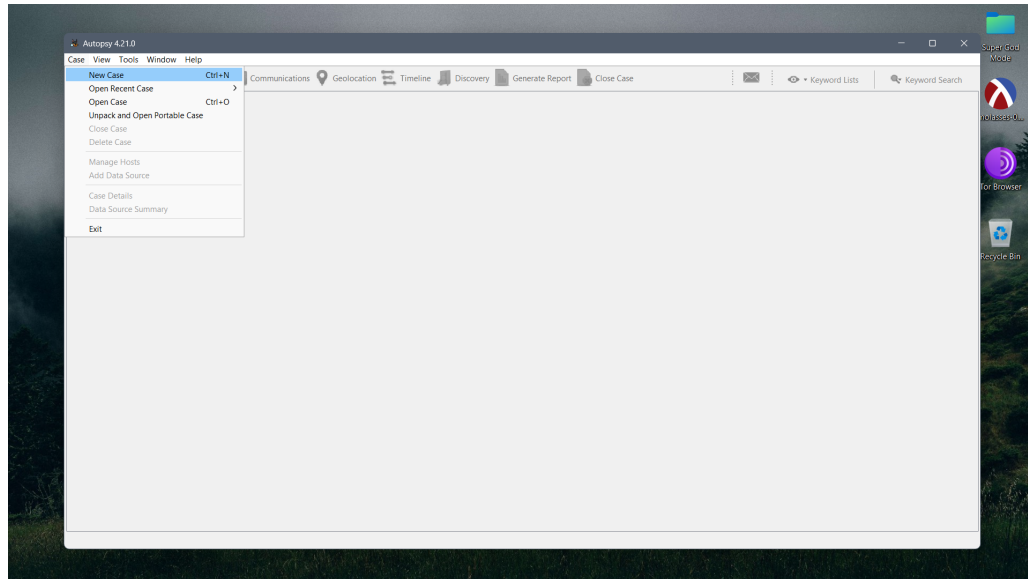


The same can be done for the Windows image as well.

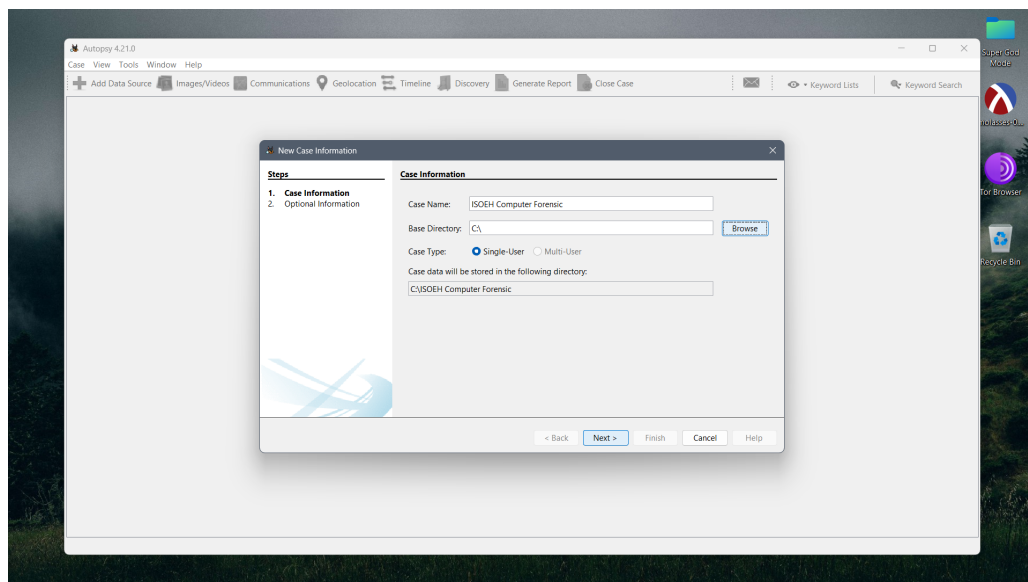


## 3.2 Methodology: Autopsy

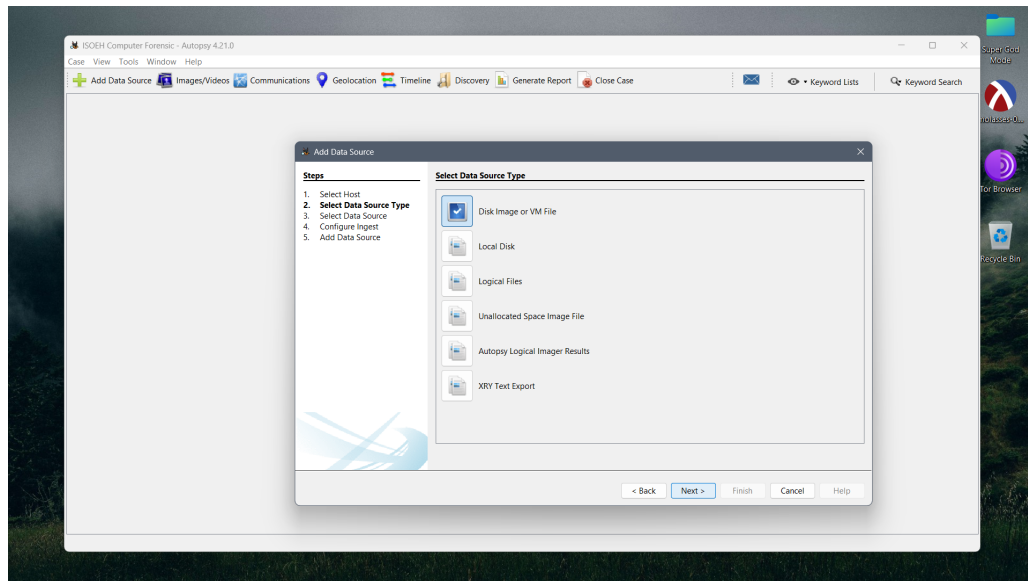
We begin by opening Autopsy, and selecting **Case > New Case**.



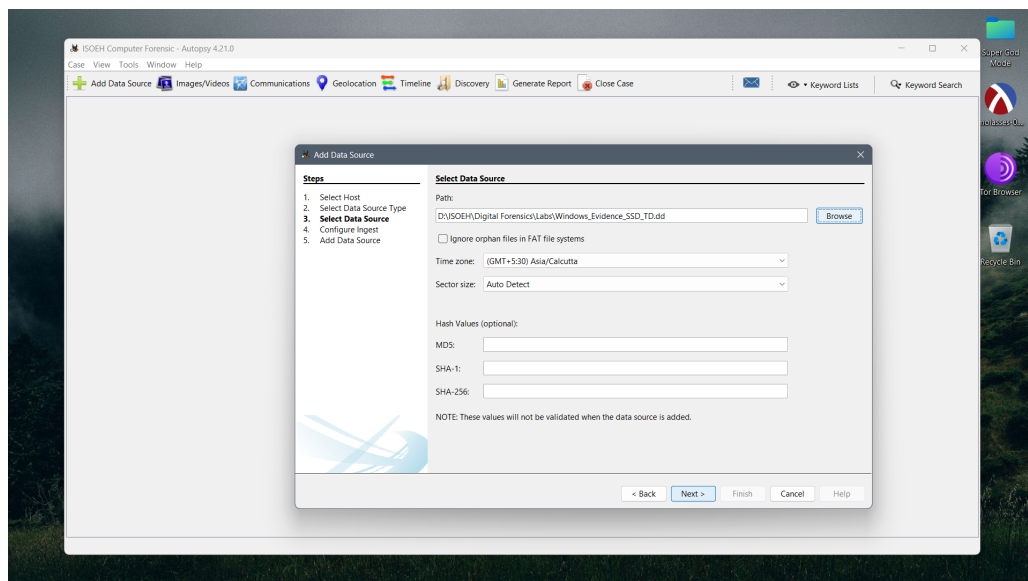
Next, we enter the information such as Case Name, Case Type (single-user or multi-user), and Examiner Information (optional), and click **Finish**.



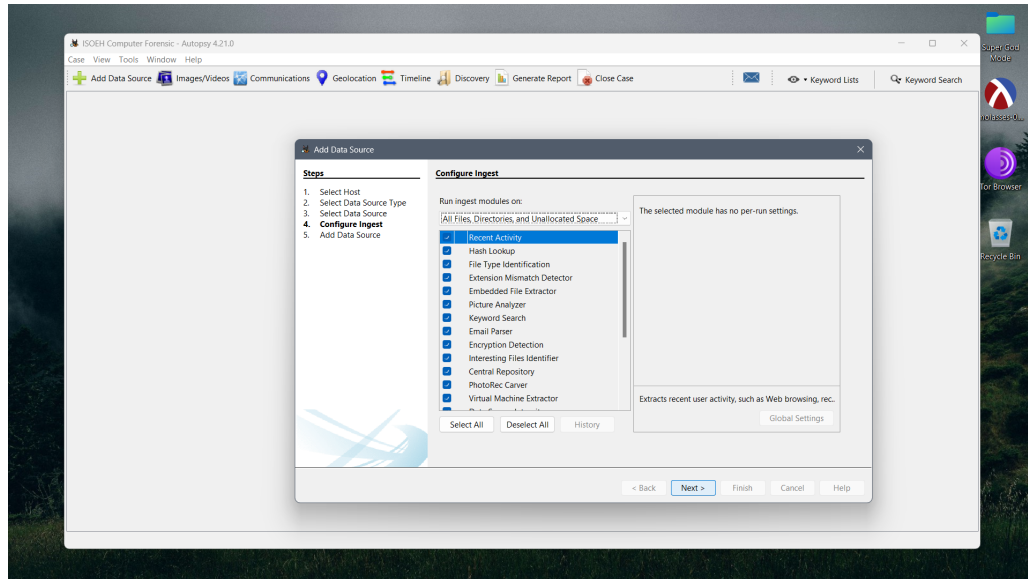
Now we need to add data sources. We select *Add Data Source* and select the data source type as *Disk Image or VM File*.



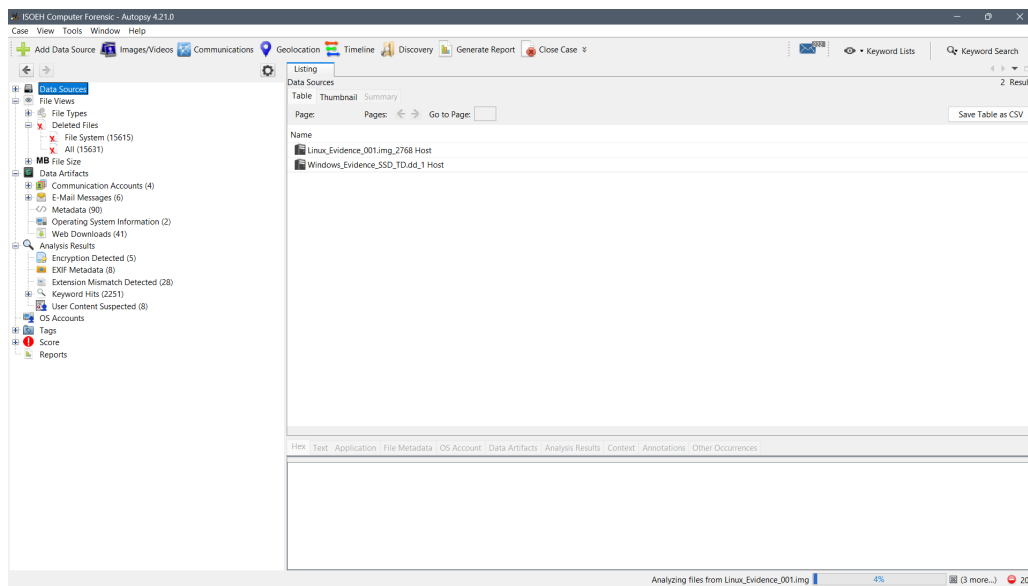
Next, we select the appropriate disk image path.



In the **Configure Ingest** page, we leave all options checked, and finish adding the data source.



Upon adding both the disk images to our case, we are ready to perform analysis.



Our analysis of the disk images shows suspicious activities such as a folder relating to Al-Qaida.

The screenshot displays the ISORH Computer Forensic - Autopsy 4.21.0 interface. The left sidebar shows a file system tree for 'Linux\_Evidence\_001.limg.2768.Host'. The 'Linux\_Evidence\_001.limg' directory is expanded, showing various subdirectories like '.cache', '.config', '.gnupg', '.local', '.mozilla', '.Desktop', '.Documents', '.Downloads', '.Log Files', '.Music', '.Pictures', '.Templates', and '.Videos'. The right pane shows the 'Listing' of files in the 'Log Files/AI-Qaida Articles' directory. The table lists files with columns: Name, S, C, O, Modified Time, Change Time, Access Time, Created Time, Size, Flags(Dr), and File Type. The file '22mar06-ikhwan.net.jpg' is highlighted.

| Name                                            | S | C | O | Modified Time           | Change Time             | Access Time             | Created Time            | Size  | Flags(Dr) | File Type         |
|-------------------------------------------------|---|---|---|-------------------------|-------------------------|-------------------------|-------------------------|-------|-----------|-------------------|
| [current folder]                                |   |   |   | 2016-06-09 16:49:55 IST | 2016-06-09 16:12:32 IST | 2016-06-09 16:12:32 IST | 2016-06-09 16:12:32 IST | 4096  | Allocated | AI-Qaida Articles |
| [parent folder]                                 |   |   |   | 2016-06-09 12:59:25 IST | 2016-06-09 16:12:32 IST | 2016-06-09 16:12:32 IST | 2016-06-09 16:12:32 IST | 4096  | Allocated | AI-Qaida Articles |
| [NoooooooooooooooooooooIAA A'oooooooooID]_files |   |   |   | 2016-06-08 16:49:55 IST | 2016-06-09 16:12:32 IST | 2016-06-09 16:12:32 IST | 2016-06-09 16:12:32 IST | 4096  | Allocated | AI-Qaida Articles |
| x_files                                         |   |   |   | 2016-06-08 16:49:55 IST | 2016-06-09 16:12:32 IST | 2016-06-09 16:12:32 IST | 2016-06-09 16:12:32 IST | 4096  | Allocated | AI-Qaida Articles |
| 22mar06-ikhwan.net.jpg                          |   |   | 1 | 2006-03-29 10:13:24 IST | 2016-06-09 16:12:32 IST | 2016-06-09 16:12:32 IST | 2016-06-09 16:12:32 IST | 31183 | Allocated | AI-Qaida Articles |
| [NoooooooooooooooooooooIAA A'oooooooooID]htm    |   |   | 1 | 2006-03-29 10:16:56 IST | 2016-06-09 16:12:32 IST | 2016-06-09 16:12:32 IST | 2016-06-09 16:12:32 IST | 718   | Allocated | AI-Qaida Articles |
| xhtm                                            |   |   | 1 | 2006-03-29 10:17:26 IST | 2016-06-09 16:12:32 IST | 2016-06-09 16:12:32 IST | 2016-06-09 16:12:32 IST | 645   | Allocated | AI-Qaida Articles |

We also find an abundance of deleted files, several of which appear to be related to malware.

YORH Computer Forensic - Autopsy 4.11.0 Case View Tools Window Help

+ Add Data Source
 Images/Videos
 Communications
 Geolocation
 Timeline
 Discovery
 Generate Report
 Close Case
 Keyword Lists
 Keyword Search

Listing

Table Thumbnail Summary

Page: 1 of 2 Pages: Go to Page: 10000 Results

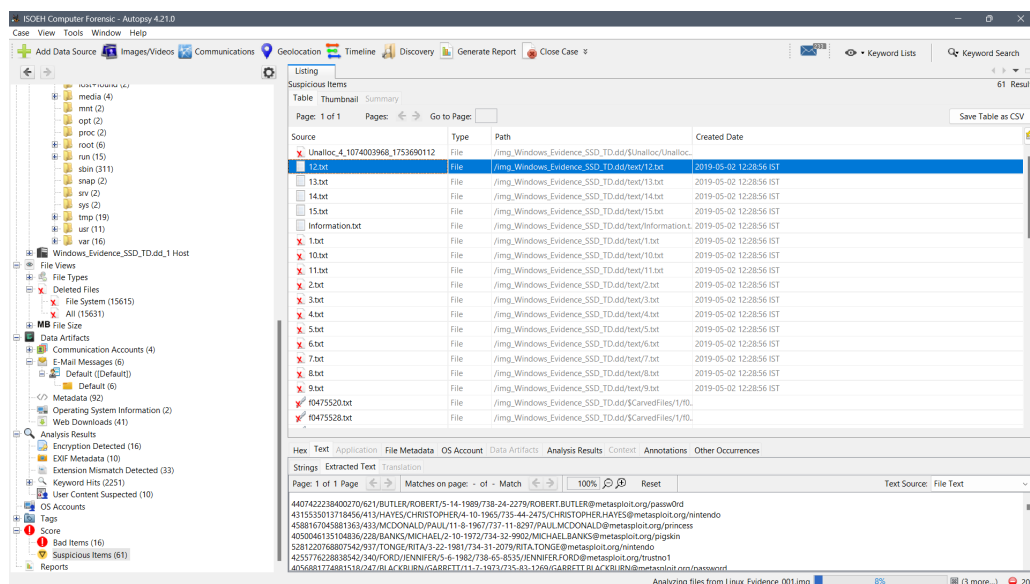
Save Table as CSV

| Name                          | S | C | O | Modified Time           | Change Time             | Access Time             | Created Time            | Size   | Flags(Drv)  |
|-------------------------------|---|---|---|-------------------------|-------------------------|-------------------------|-------------------------|--------|-------------|
| ✖ 0475816.gif                 |   | 0 | 0 | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 29006  | Unallocated |
| ✖ 0475952.gif                 |   | 0 | 0 | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 34445  | Unallocated |
| ✖ 0476024.jpg                 |   | 0 | 0 | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 10953  | Unallocated |
| ✖ 0476048.png                 |   | 0 | 0 | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 5701   | Unallocated |
| ✖ 0476064.jpg                 |   | 0 | 0 | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 5691   | Unallocated |
| ✖ unity-settings-diamond.log  |   | 0 | 0 | 2016-06-09 16:08:25 IST | 2016-06-09 16:09:21 IST | 2016-06-09 16:08:23 IST | 2016-06-09 16:08:23 IST | 565    | Unallocated |
| ✖ registry_HQURYI             |   | 0 | 0 | 2016-06-09 15:10:16 IST | 2016-06-09 15:10:16 IST | 2016-06-09 16:05:26 IST | 2016-06-09 15:10:16 IST | 12921  | Unallocated |
| ✖ registryadb_64.bin.tmp11Z7Y |   | 0 | 0 | 2016-06-09 15:10:20 IST | 2016-06-09 15:10:21 IST | 2016-06-09 15:48:18 IST | 2016-06-09 15:10:20 IST | 273997 | Unallocated |
| ✖ firmware.xml.gz.PLLVf       |   | 0 | 0 | 2016-06-09 15:11:28 IST | 2016-06-09 15:11:28 IST | 2016-06-09 15:11:28 IST | 2016-06-09 15:11:28 IST | 4720   | Unallocated |
| ✖ status.mp                   |   | 0 | 0 | 2016-06-09 16:09:22 IST | 2016-06-09 16:09:22 IST | 2016-06-09 16:09:23 IST | 2016-06-09 16:09:23 IST | 872    | Unallocated |
| ✖ test-unwanted-simplestore   |   | 0 | 0 | 2016-06-09 16:12:26 IST | 2016-06-09 16:12:26 IST | 2016-06-09 16:12:26 IST | 2016-06-09 16:12:26 IST | 239    | Unallocated |
| ✖ test-track-simple.pst       |   | 0 | 0 | 2016-06-09 16:12:07 IST | 2016-06-09 16:12:07 IST | 2016-06-09 16:11:51 IST | 2016-06-09 16:11:51 IST | 4096   | Unallocated |
| ✖ test-malware-simple.cache   |   | 0 | 0 | 2016-06-09 16:11:51 IST | 2016-06-09 16:11:51 IST | 2016-06-09 16:11:51 IST | 2016-06-09 16:11:51 IST | 4096   | Unallocated |
| ✖ test-malware-simple.pst     |   | 0 | 0 | 2016-06-09 16:12:55 IST | 2016-06-09 16:12:55 IST | 2016-06-09 16:11:51 IST | 2016-06-09 16:11:51 IST | 4096   | Unallocated |
| ✖ test-phin-simple.cache      |   | 0 | 0 | 2016-06-09 16:11:51 IST | 2016-06-09 16:11:51 IST | 2016-06-09 16:11:51 IST | 2016-06-09 16:11:51 IST | 244    | Unallocated |
| ✖ test-track-simple.store     |   | 0 | 0 | 2016-06-09 16:11:51 IST | 2016-06-09 16:11:51 IST | 2016-06-09 16:11:51 IST | 2016-06-09 16:11:51 IST | 260    | Unallocated |
| ✖ test-unwanted-simple.cache  |   | 0 | 0 | 2016-06-09 16:11:51 IST | 2016-06-09 16:11:51 IST | 2016-06-09 16:11:51 IST | 2016-06-09 16:11:51 IST | 259    | Unallocated |
| ✖ test-malware-simple.store   |   | 0 | 0 | 2016-06-09 16:11:51 IST | 2016-06-09 16:11:51 IST | 2016-06-09 16:11:51 IST | 2016-06-09 16:11:51 IST | 270    | Unallocated |

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Analzing files from Linux Evidence 001.img 8% (3 more...) 100%

In fact, we discover a huge dump of several passwords obtained through the use of Metasploit Framework.



Our investigations lead us to conclude that the user is involved in several illegal activities. As such, we can gather the evidence and proceed.